

Presentation of a paper:

The Economic Costs of Conflict: A Case Study of the Basque Country
A. Abadie & J. Gardeazabal (2001)

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Contents

1 Introduction

- Authors
- Some context

2 Research

- Scenario Model of GDP
- Stocks Modeling

3 Q&A

Background of the authors

Academic

Alberto Abadie

currently a professor of Economics at MIT, at the time of writing assistant professor at Harvard.

Javier Gardeazabal

currently a professor at the University of the Basque Country, Phd at the University of Pennsylvania in May 1991.

Basque during Franco and post Franco

Franco Basque

Under Franco, the Basque Country suffered political repression and centralized economic policies that constrained regional autonomy and industrial investment, reducing long-term growth potential. Despite this the Basque region in the 1960s got one of the highest GDP per Capita, investment ration (investment/production), density of population, educated labor force in the whole of Spain.

post Franco Basque:

After Franco, democratization and restored regional authority allowed economic recovery and stronger institutions, but violence from ETA continued to impose measurable economic losses.

Basque during Franco and post Franco

Franco Basque



Post-Franco Basque



How the Authors simulated the Counterfactual scenario?

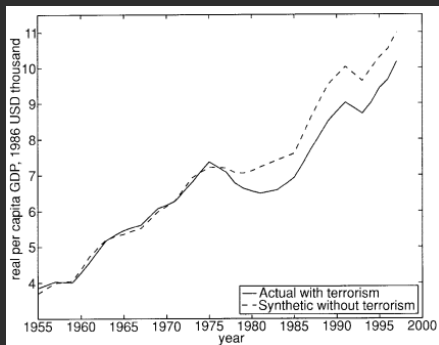
- They compared the Basque Country's observed economic path to a weighted combination of the 16 other Spanish regions to construct a “synthetic” Basque Country without terrorism.
- Choose nonnegative weights W that sum to one so the weighted average of control regions best matches Basque pre-terrorism characteristics. Where V is a diagonal matrix weighting predictor of importance chosen empirically.

$$(X_1 - X_0W)'V(X_1 - X_0W)$$

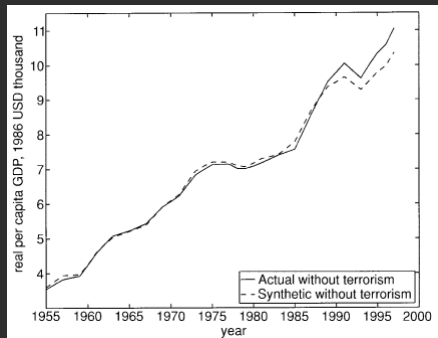
- The resulting W^* uses Catalonia (0.8508) and Madrid (0.1492) as positive-weight controls; others get zero weight. Assuming that Basque kept up with the growth (Catalonia and Madrid retained) it had during Franco.

Synthetic & Placebo

Synthetic



Catalonia Placebo



Fama-French Models

At the time of writing the 3 variable one was existent. The five model one was updated in 2015.

$$r = R_f + \beta(R_m - R_f) + b_s \cdot SMB + b_v \cdot HML + b_p \cdot RMW_5 + b_i \cdot CMA_5 + \alpha$$

R_f - Risk free return rate

R_m - Return of the market portfolio

SMB - Small market cap. - Big market cap. (returns)

HML - High B/M - Low B/M (returns)

RMW_5 - High Operating prof. - Low Operating prof. (returns)

CMA_5 - Conservative investing - Intensive investing (returns)

In the US (1963–2013), adding these two factors makes HML redundant: its returns are explained by the other four factors, especially CMA (corr ≈ 0.7).

Natural Experiment using the Truce of 1998-1999

The authors used in total 3 types of model to test out the causality of the effects of EPAs activity during the temporary truce.

French-Fama

$$R_t^j = \alpha^j + \beta_1^j R_t^m + \beta_2^j SMB_t + \beta_3^j HML_t + AR_t^j$$

Noise-less day after the truce portfolio

$$CAR_t^j = \left(\prod_{s=1}^t \{1 + AR_s^j\} \right) - 1$$

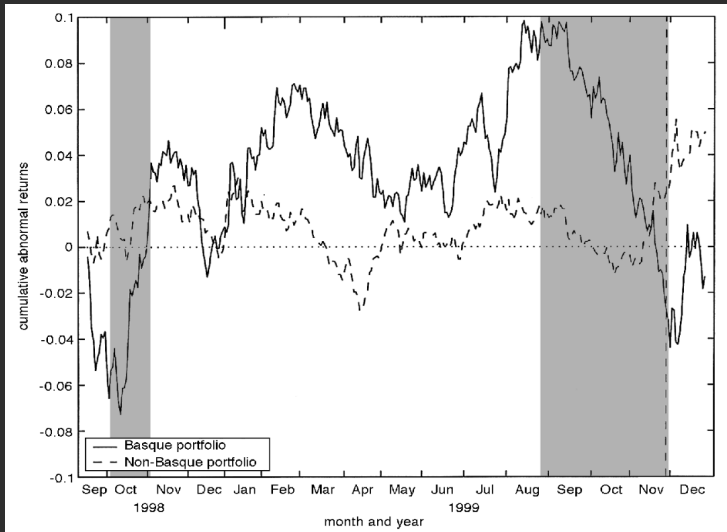
Market Model

$$R_t^j = \gamma^j + \lambda R_t^m + AR_t^j$$

Constant Mean Return

$$R_t^j = \mu^j + AR_t^j$$

Noise-less day after truce portfolio comparison



Comparison of coefficients of the three main models

	Market Model			Constant-Mean-Return Model			Fama-French Three-Factor Model plus term and risk structure factors		
	Basque (1)	Non-Basque (2)	Difference (3)	Basque (4)	Non-Basque (5)	Difference (6)	Basque (7)	Non-Basque (8)	Difference (9)
Constant	-0.0004 (0.0003)	0.0001 (0.0002)	-0.0004 (0.0004)	-0.0003 (0.0004)	0.0001 (0.0006)	-0.0004 (0.0005)	-0.0006 (0.0009)	0.0000 (0.0004)	-0.0006 (0.0009)
R^m	0.4696 (0.0322)	0.9312 (0.0150)	-0.4617 (0.0375)				0.6726 (0.0362)	0.8083 (0.0187)	-0.1356 (0.0345)
SMB							0.3645 (0.0461)	-0.2272 (0.0237)	0.5917 (0.0445)
HML							0.2553 (0.0398)	-0.1412 (0.0208)	0.3964 (0.0382)
$TERM$							0.0262 (0.0478)	0.0247 (0.0258)	0.0015 (0.0452)
DEF							-0.0341 (0.1643)	-0.0823 (0.0917)	0.0482 (0.1577)
Good News	0.0054 (0.0023)	0.0002 (0.0010)	0.0052 (0.0026)	0.0100 (0.0033)	0.0094 (0.0039)	0.0006 (0.0028)	0.0050 (0.0022)	0.0006 (0.0009)	0.0044 (0.0022)
Bad News	-0.0021 (0.0008)	0.0004 (0.0005)	-0.0025 (0.0009)	-0.0017 (0.0010)	0.0013 (0.0015)	-0.0030 (0.0012)	-0.0020 (0.0011)	-0.0002 (0.0006)	-0.0018 (0.0011)
R^2	0.3939	0.8887	0.3195	0.0257	0.0127	0.0053	0.4969	0.9109	0.5500

Notes: Heteroskedasticity-robust standard errors are in parentheses. The sample period consists of 747 trading sessions for 1998–2000.

Ending remarks

I am delighted for your attention.

Q & A?